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■ OBJECTIVES

After completing this chapter, the reader should be able to:

- Describe the principle of operation of the image intensifier.
- State how exposure rate varies when the field of view is changed on an image intensifier.
- Explain the principle of operation of television.
- Determine the vertical and horizontal resolution of a television system.
- Explain how image processing can be used to improve signal to noise ratio in digital fluoroscopy.
- Name the events that take place when digital subtraction angiography is performed.
- Give two methods that are used to remove motion artifacts from digital subtraction angiography images.
- Describe how photospot images are obtained.
- Trace the chain of events that take place when an automatic brightness control system varies exposure rate.

Information about moving structures within a patient may be obtained with radiographs exposed in rapid succession. However, the interval of time between serially exposed films may be too great to provide complete information about dynamic processes within the body. Consequently, a technique is required to furnish images that reflect near instantaneous changes occurring in the patient. This technique is referred to as *fluoroscopy*.

■ FLUOROSCOPY AND IMAGE INTENSIFICATION

In early fluoroscopic techniques, x rays emerging from the patient impinged directly on a fluoroscopic screen. Light was emitted from each region of the screen in response to the rate at which energy was deposited by the incident x rays. The light image on the fluoroscopic screen was viewed by the radiologist from a distance of 10 or 15 in. A thin plate of lead glass on the back of the fluoroscopic screen shielded the radiologist from x radiation transmitted by the screen.

Using this fluoroscopic technique, the radiologist perceived a very dim image with poor visibility of detail. Radiologists had to “dark adapt” their eyes by remaining in the dark for extended periods in order to view the images. Radiologists recognized in the 1940s that the poor visibility of image detail in fluoroscopy was related to the dim image presented by early fluoroscopes.^{1,2} They emphasized the need for brighter fluoroscopic images and encouraged the development of the *image intensifier*. Image intensifiers increase the brightness of the fluoroscopic image and permit the observer to use photopic (cone) vision in place of the scotopic (rod) vision required with earlier fluoroscopes. Because of the brighter images, dark adaptation is not required for fluoroscopy with image intensification. Although the image intensifier has increased the cost and complexity of fluoroscopic systems, the use of non-image-intensified fluoroscopy has been outmoded for some time.

X-Ray Image Intensifier Tubes

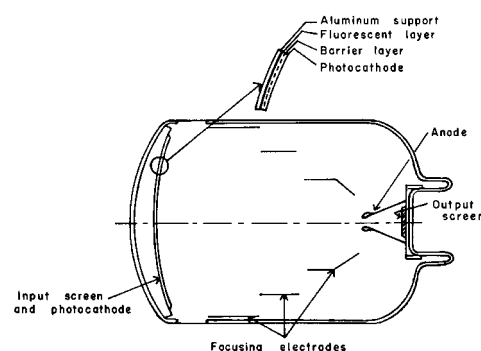
The x-ray image intensifier “intensifies” or increases the brightness of an image through two processes: (1) minification, in which a given number of light photons emanates from a smaller area, and (2) flux gain, where electrons accelerated by high voltages produce more light as they strike a fluorescent screen. This same principle is employed in “night vision” devices used by the military to observe objects in low light conditions.

An image intensifier tube is diagramed in Margin Figure 14-1. X rays impinge upon a fluorescent screen (input screen) that is from 4 inches to greater than 16 inches in diameter and slightly convex in shape. The fluorescent emulsion is a thin layer of CsI (cesium iodide). Older image intensifier input screens were composed of ZnS: CdS (zinc cadmium sulfide). The principal advantage of CsI over ZnS: CdS: Ag is the increased absorption of x rays because of the presence of higher-Z components in the CsI phosphor and because of the increased packing density of CsI molecules in the phosphor granules.

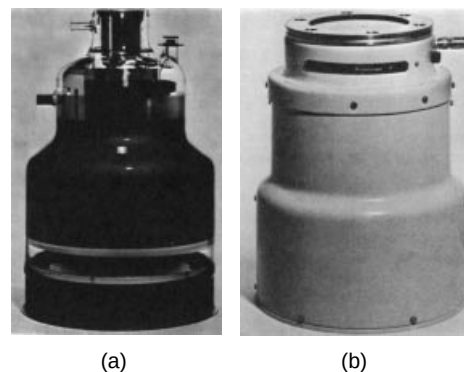
For each x-ray photon absorbed, 2000 to 3000 photons of light are emitted by the screen. These light photons are not observed directly. Instead, the light falls on a photocathode containing the element antimony (Sb), such as antimony cesium oxide (Sb-CsO).³ Light photons released in a direction away from the photocathode are reflected toward the photocathode by a mirrored aluminum support on the outside surface of the input screen. If the spectral sensitivity of the photocathode is matched to the wavelength of light emitted by the screen, then 15 to 20 electrons are ejected from the photocathode for every 100 photons of light received. The number of electrons released from any region of the photocathode depends upon the number of light photons incident upon the region. The electrons are accelerated through a potential difference of 25 to 35 kV between the photocathode and the anode of the image intensifier tube. The electrons pass through a large hole in the anode and strike a small fluorescent screen (output screen) mounted on a flat glass support. The emulsion on the output screen resembles that for the input screen, except that the fluorescent granules are much smaller. Diameters of most output screens range from $\frac{1}{2}$ in. to 1 in. Intensifiers with small output screens are used for television fluoroscopy because the diameter of the input screen of a television camera is small also. A coating of metal, usually aluminum, is deposited on the output screen to prevent the entrance of light from outside the intensifier. The metallic layer also removes electrons accumulated by the output screen.

Electrons from the photocathode are focused on the output screen by cylindrical electrodes positioned between the photocathode and the anode. Usually, three focusing electrodes are used (Margin Figure 14-2). The glass envelope is contained within a housing of Mumetal, an alloy containing iron. The housing attenuates magnetic fields that originate outside the intensifier and prevents these fields from distorting the motion of electrons inside. The motion of electrons and, therefore, the image on the output phosphor may still be distorted by a strong magnet field around the intensifier. Also, an intense magnetic field in the vicinity of the image intensifier may magnetize the Mumetal shield and focusing electrodes and cause permanent distortion of the fluoroscopic image. Consequently, image intensifiers should not be located near permanent or transient magnetic fields of high intensity as may be found near magnetic resonance imaging systems (see Chapter 25).

With an x-ray image intensifier, four different information carriers transmit information about the patient to the radiologist. The x-ray beam transmits information from the patient to the input screen of the image intensifier. At the input screen, the information carrier is changed from x rays to photons of visible light. As the light photons are absorbed by the photocathode, the information is transferred to an electron beam, which is directed upon the output screen of the intensifier. The information is transmitted as a light image from the output screen to the observer's retina.



MARGIN FIGURE 14-1
Cross section of a typical x-ray image intensifier tube.



MARGIN FIGURE 14-2
A: Image intensifier tube with an input screen 9 in. in diameter. B: Tube encased in its housing.

Gain and Conversion Efficiency of Image Intensifiers

The brightness of the image on the output screen of an image intensifier may be compared with the brightness of the image provided by a standard non-image-intensified fluoroscopic screen. The image-intensifier and the fluoroscopic screen receive identical exposures to radiation, and the ratio of the brightness of the two images is termed the *brightness gain* of the image intensifier.

$$\text{Brightness gain} = \frac{\text{Brightness of the output screen of the image intensifier}}{\text{Brightness of a standard screen}} \quad (14-1)$$

The brightness gain of image intensifiers varies from 1000 to over 6000, depending on the particular image intensifier used and the fluorescent screen with which it is compared. The gain in brightness results from two independent processes that occur within the intensifier. These processes are *image minification* and *flux gain*.

The light image produced as x rays are absorbed in the input screen of an image intensifier is reproduced as a minified image on the output screen of the intensifier. Because the output screen is much smaller than the input screen, the amount of light per unit area from the output screen is greater than that from the input screen. The increase in image brightness furnished by minification of the image is referred to as the *minification gain* g_m and is equal to the ratio of the areas of the input and output screens:

$$\begin{aligned} g_m &= \frac{\text{Area of input screen}}{\text{Area of output screen}} \\ &= \frac{\pi[(\text{diameter of input screen})/2]^2}{\pi[(\text{diameter of output screen})/2]^2} \\ &= \frac{(\text{Diameter of input screen})^2}{(\text{Diameter of output screen})^2} \end{aligned} \quad (14-2)$$

For example, the minification gain is 81 for an image intensifier with an input screen 9 in. in diameter and an output screen 1 in. in diameter:

$$g_m = \frac{(9 \text{ in.})^2}{(1 \text{ in.})^2}$$

The brightness of the image on the output screen is also increased because electrons from the photocathode are accelerated as they travel toward the output screen. As these electrons are stopped in the output screen, the number of light photons released varies with the energy of the electrons. The gain in brightness due to electron acceleration is termed the *flux gain* g_f of the image intensifier. A typical image intensifier has a flux gain of at least 50.

The *total gain* g in luminance of an image intensifier is the product of the minification gain g_m and the flux gain g_f :

$$g = (g_m)(g_f) \quad (14-3)$$

For example, the luminance gain g is 4050 for an image intensifier with a minification gain of 81 and a flux gain of 50:

$$\begin{aligned} g &= (81)(50) \\ &= 4050 \end{aligned}$$

Two image intensifiers may be compared by describing the conversion factor for each intensifier.⁴ The conversion factor G_r is the quotient of the luminance of the output screen of the image intensifier divided by the exposure rate at the input screen. The luminance is expressed in units of candela per square meter and the exposure rate in milliroentgens per second:

$$G_r = \frac{\text{Luminance of output screen (candela/m}^2\text{)}}{\text{Exposure rate at input screen (mR/sec)}} \quad (14-4)$$

The conversion factor for an image intensifier is dependent on the energy of the radiation and should be measured with x radiation from a full-wave-rectified or constant-potential x-ray generator operated at about 85 kVp. The conversion factor of most image intensifiers ranges from 50 to 100 (candela-sec)/(mR-m²).

The luminance or “brightness” of an object is described in units of lamberts. One lambert (L) is the luminance of a perfectly diffusing surface that is emitting or reflecting 1 lumen/cm²; 1 mL = 0.001 L.

The candela is a unit of luminance and equals 1/60 of the luminance of a square centimeter of a blackbody heated to the temperature of solidifying platinum (1773.5° C). One candela is equivalent to 0.3 millilamberts.

Resolution and Image Distortion with Image Intensifiers

The resolution of an image intensifier is limited by the resolution of the input and output fluorescent screens and by the ability of the focusing electrodes to preserve the image as it is transferred from the input screen to the output screen. The resolution of image intensifiers averages about 4 lines per millimeter for intensifiers with CsI input screens. Contributions to resolution loss that originate outside the image intensifier include the presence of scattered radiation in the x-ray beam received by the input screen and unsharpness in the image contributed by patient motion and the finite size of the focal spot. In addition, the quality of the fluoroscopic image is affected by statistical fluctuations in the number of x rays impinging on the input screen.

The resolution, brightness, and contrast of an image provided by an image intensifier are greatest in the center of the image and reduced toward the periphery. The decrease in brightness along the periphery is usually less than 25%. The reduction in brightness and image quality along the border of the fluoroscopic image is referred to as *vignetting*. Vignetting is a reflection of (a) the reduced exposure rate along the periphery of the input screen and (b) the reduced precision with which electrons from the periphery of the photocathode strike the output screen. Also, the center of the output phosphor receives some scattered light from surrounding regions of the output phosphor, whereas the periphery receives contributions only from the center. Therefore, the lack of a contribution of light from the area beyond the output screen also contributes to vignetting.

Straight lines in an object often appear to curve outward in the fluoroscopic image. This effect is referred to as *pincushion distortion*, caused by the curvature of the input screen and by the reduced precision with which electrons from the periphery of the photocathode are focused upon the output screen. Image lag of the input and output screens of an image intensifier may occur if the phosphor is slow to respond to changes in image brightness. This effect may be of concern during procedures such as high-frame-rate cinefluorography and fast digital imaging.

Size of Image Intensifier

The diameter of the input screen of an image intensifier ranges from 4 to more than 16 in. Intensifiers with small input screens are more maneuverable and less expensive. A small intensifier furnishes a slight improvement in resolution because electrons from the photocathode strike the output screen with greater precision. However, the region of the patient encompassed by the input screen is restricted with a small intensifier. A larger intensifier is more expensive and less maneuverable but furnishes a larger field-of-view and greater opportunity for image magnification.

The diameter of the input screen of an image intensifier is larger than the diameter of the region studied within the patient. In Margin Figure 14-3, for example, an object of length s has a length s' in the image on the input screen. If d is the distance from the target of the x-ray tube to the object and d' is the distance from the target to the input screen, then the apparent length s' of an object of actual length s is

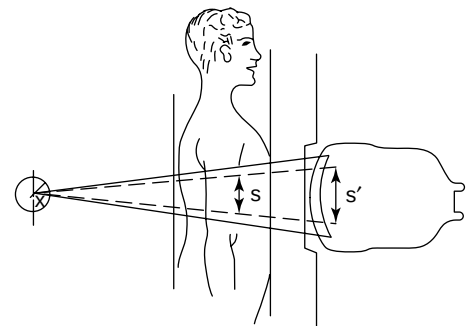
$$s' = s \left(\frac{d'}{d} \right) \quad (14-5)$$

The magnification M of the image is given by

$$M = \frac{s'}{s} = \frac{d'}{d} \quad (14-6)$$

Example 14-1

An x-ray tube is positioned 45 cm below a fluoroscopic table. The input screen of an image intensifier is 15 cm in diameter and 30 cm above the table. What is the



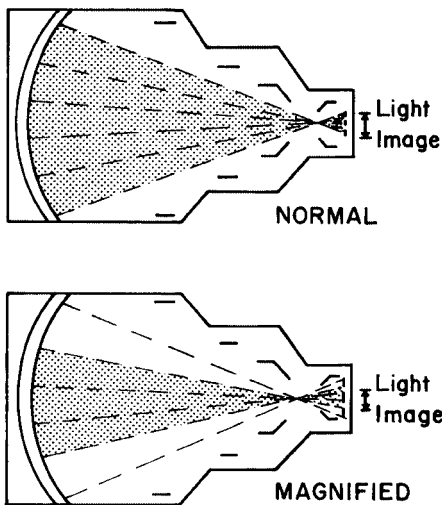
MARGIN FIGURE 14-3

An object of length s has an apparent length s' on the input screen of an image intensifier. The field of view of an image intensifier is smaller than the size of the input screen.

maximum length s of an object that is included completely on the screen? What is the magnification of the image? The object is 10 cm above the table.

$$s' = s \left(\frac{d'}{d} \right) \quad (14-5)$$

$$\begin{aligned} s &= s' \left(\frac{d}{d'} \right) \\ &= (15 \text{ cm}) \left(\frac{10 \text{ cm} + 45 \text{ cm}}{30 \text{ cm} + 45 \text{ cm}} \right) \\ &= 11 \text{ cm} \\ M &= \frac{s'}{s} \\ &= \frac{15 \text{ cm}}{11 \text{ cm}} \\ &= 1.36 \end{aligned} \quad (14-6)$$



MARGIN FIGURE 14-4

Dual-field image intensifier. By changing the voltage on the focusing electrodes, two different electron crossover points are produced that correspond to normal and magnified modes of viewing.

Dual- and Triple-Field Intensifiers

Many image intensifiers permit magnified viewing of the central region of the input screen. These intensifiers are termed *dual-field* intensifiers if one mode of magnified viewing is provided, and they are called *triple-field* intensifiers if two magnified viewing modes are offered. The operation of a dual-field intensifier is diagramed in Margin Figure 14-4. In the normal viewing mode, electrons from the photocathode converge upon the electron crossover point at the location nearest the output screen and strike only the region of the output screen visible to the TV camera. In this manner, the observer views the entire input screen of the intensifier. When the intensifier is operated in magnified mode, the voltage on the focusing electrodes is changed, and electrons converge upon a crossover point farther from the output screen. Under this condition, electrons originating from the periphery of the input screen strike the output screen outside the region viewed by the TV camera. Only the central region of the input screen is viewed, and this region is seen as a magnified image. In a 22.5-cm (9-in.) dual-field image intensifier, only the central 15 cm (6 in.) of the input screen are seen in the magnified viewing mode.

If the voltage on the focusing electrodes is changed again, electrons from the photocathode can be forced to converge upon a crossover point even farther from the output screen. In this manner, a second mode of magnified viewing is presented to the observer. An intensifier with two magnified viewing modes is termed a *triple-field image intensifier*. Many 22.5-cm triple-field image intensifiers provide an image of the entire input screen in the normal viewing mode and an image of the central 15 cm (6 in.) and 11 cm (4.5 in.) of the input screen during magnified viewing.

In the magnified viewing mode, the image on the output screen is produced by electrons from only the central portion of the input screen. Because fewer electrons are used to produce the image, the brightness of the image would be reduced unless the exposure rate to the input screen is increased. This increase in exposure rate is accomplished automatically as the viewing mode is switched from normal to magnified. The exposure rate increases with decreasing area of the active input phosphor.

Example 14-2

Estimate the increase in exposure rate required to maintain image brightness when an image intensifier field of view is decreased from 9 in. to 6 in. in diameter.

Because the brightness of the image depends on the area of the image intensifier, the increase is determined by the ratio of the square of the two diameters.

$$\frac{\text{Exposure rate at 6 in.}}{\text{Exposure rate at 9 in.}} = \frac{(9 \text{ in.})^2}{(6 \text{ in.})^2} = 2.25$$

Thus the exposure is increased by a factor of 2.25 when the field of view is decreased from 9 to 6 in. if no other adjustments are made.

TELEVISION DISPLAY OF THE FLUOROSCOPIC IMAGE

The image furnished by an image intensifier may be televised by viewing the intensifier output screen with television camera. The signal from the camera is transmitted by cable to a television monitor. This method for transmission and display of an image is referred to as *closed-circuit television* because the signal is transmitted from the television camera to the monitor by a coaxial cable rather than through air (Margin Figure 14-5).

Television Cameras

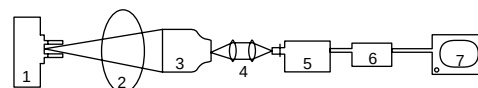
Television cameras, sometimes referred to as television pickup tubes, average about 15 cm in length and 3 cm in diameter (Margin Figure 14-6). In a typical analog camera a thin layer of photoconducting material (usually antimony trisulfide [SbS_3] in the vidicon camera) is plated on the inner surface of the face of a glass envelope and is separated from the glass by a thin, transparent conducting layer. The side of the photoconducting layer nearest to the face of the glass envelope is 20 to 60 V positive with respect to the cathode of the electron gun. The photoconducting layer is an electrical insulator when not exposed to light. When light from the output screen of the image intensifier falls upon the photoconducting layer, the resistance across the layer decreases. That is, the resistance across any region of an illuminated photoconducting layer depends on the amount of light incident on the region.

The scanning electron beam from the electron gun deposits electrons on the photoconducting layer. Some of the electrons migrate toward the positive side of the layer and are collected by the conducting layer. In any region, the number of electrons that migrate to the positive side depends on the resistance and, consequently, the illumination of the region. Electrons collected by the conducting layer furnish an electrical signal that fluctuates in response to variations in the illumination of the photoconducting layer. This signal is amplified and used to control the intensity of the electron beam that is scanned across the fluorescent screen of the television monitor in synchrony with the scanning electron beam in the camera. Hence, the image on the television monitor corresponds closely to the image on the output screen of the image intensifier.⁵

In charge coupled device (CCD) cameras (see Chapter 8 for information on CCD detectors), the photoconduction and scanning are replaced with arrays of CCDs that produce electrical signals in response to the amount of visible light falling upon them. These electrical signals may be stored as arrays of digital values or may be converted directly to standard television signals. As in the consumer market, CCD cameras are replacing analog cameras in newer fluoroscopic systems.

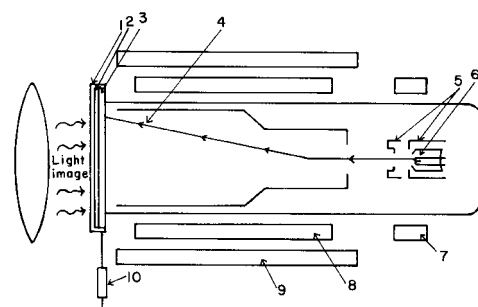
Television Scanning of a Fluoroscopic Image

As the scanning electron beam moves horizontally across the photoconducting layer of the television camera (or as CCD elements are struck by photons), the signal from the conducting layer is modulated in response to the illumination of each photoconducting element in the path of the scanning electron beam. This motion of the scanning electron beam is termed an *active sweep*. As the electron beam is scanned horizontally in the opposite direction (*retrace sweep*), no information is transmitted to the television monitor. The retrace sweep is about five times faster than the active



MARGIN FIGURE 14-5

Typical system for television fluoroscopy: 1, x-ray tube; 2, patient; 3, image intensifier; 4, optical system; 5, television camera; 6, camera control unit; 7, television monitor.



MARGIN FIGURE 14-6

Vidicon television pickup tube: 1, glass envelope; 2, transparent conducting layer; 3, photoconducting layer; 4, scanning electron beam; 5, grids; 6, electron gun; 7, alignment coil for the scanning beam; 8, horizontal and vertical deflection coils; 9, focusing coils; 10, load resistor.

sweep. As the next active sweep is initiated, it is displaced slightly below the preceding sweep. The number of active sweeps of the scanning electron beam across the image may be 525, 625, 837, 875, 945, or 1024. Hence the image is divided into 525, 625, 837, 875, 945 or 1024 horizontal lines on the screen of the television monitor. (The actual number of lines that appear is slightly smaller because of the vertical retrace; see below.) Images with a large number of horizontal scan lines provide better vertical resolution than those formed with a smaller number of lines.⁶

To prevent brightness flicker in the image, the electron beam of the television camera first scans the photoconducting layer along even-numbered sweeps (2, 4, 6, 8, ...). This operation requires 1/60 second in standard television systems. The resulting set of horizontal scan lines or *field* is then interlaced with sweeps of the electron beam along odd-numbered lines (1, 3, 5, 7, ...). The final image, produced by two interlaced fields, is referred to as a *frame*. A frame is completed in 1/30 second. If the two “half images” or fields were not interlaced (if the electron beam swept sequentially along all lines (1, 2, 3, 4, ...)), the image would appear at a rate of 30 times per second and a sensation of flickering light would be perceived. An alternative is the use of special high frame rate monitors that present full frames 60–85 times per second.

After the entire photoconducting layer has been scanned once to compose a television field, the electron beam is returned to the top of the image to begin the next set of interlaced scan lines. The return of the scanning beam requires about 10^{-4} seconds and is referred to as the *vertical retrace*. Because of the time required for the vertical retrace, not all of the scan lines theoretically available are realized in practice. In a television system with 525 possible lines, for example, the image is actually formed with about 490 horizontal lines.⁷ The remaining 35 lines are lost during the time required for the vertical retrace.

The vertical resolution of a television image is determined primarily by the number of active sweeps of the electron beam across the anode or photoconducting layer and by the *Kell factor*. The Kell factor is the fraction of the active sweeps that are actually effective in preserving detail in the image. The Kell factor is about 0.7 for most television systems.⁸

$$\text{Vertical resolution} = (\text{Number of active sweeps/frame})(\text{Kell factor}) \quad (14-7)$$

If the number of active sweeps is increased, then the visibility of detail may be improved in the vertical direction across the television image. Of course, a camera with more active sweeps will not improve the vertical resolution if the detail is not available initially in the image on the output screen of the image intensifier.

The horizontal resolution of a television image is determined primarily by the number of resolution elements available across a single horizontal scan line. The number of resolution elements available depends primarily upon the number of scan lines per frame and upon the range of frequencies transmitted by the electronic circuitry between the television camera and monitor. The highest frequency of signal modulation that is transmitted undistorted from the television camera to the television monitor is referred to as the *frequency bandpass* or *bandwidth* of the television system. Bandwidths for closed-circuit television systems used in radiology range from 3.5 MHz (3.5 megacycles per second) to 12 MHz. A bandwidth of 3.5 MHz is used in broadcast television. The influence of bandwidth in the resolution of the television image is illustrated in the simplified problem in Example 14-3.

Example 14-3

A 525-line television system provides about 490 active sweeps per frame and a vertical resolution of

$$\begin{aligned} \text{Vertical resolution} &= (\text{Number of active sweeps/frame})(\text{Kell factor}) \\ &= (490 \text{ active sweeps/frame})(0.7) \\ &= 343 \text{ lines} \end{aligned} \quad (14-7)$$

What minimum bandwidth is required to provide equal horizontal (343 resolution elements per line) and vertical (343 lines per frame) resolution in the image?

$$\begin{aligned} \text{Number of resolution elements transmitted per second} &= (343 \text{ resolution elements/line})(343 \text{ lines/frame}) \\ &\times (30 \text{ frames/sec}) \\ &= 3.53 \times 10^6 \text{ resolution elements/sec} \end{aligned}$$

To transmit this number of resolution elements per second, a television system is required with a bandwidth of at least 3.53×10^6 Hz. That is, a system bandwidth of 3.5 MHz is barely adequate for this application and may sacrifice some image detail for very closely spaced structures in the object.

A more exact expression for the horizontal resolution in resolution elements per line is⁷

$$\text{Horizontal resolution} = \frac{2 [\text{bandwidth}] \left[\frac{\text{horizontal}}{\text{fraction}} \right] \left[\frac{\text{aspect}}{\text{ratio}} \right]}{(\text{frames/sec})(\text{lines/frame})} \quad (14-8)$$

In Equation 14-8, (1) the bandwidth is expressed in hertz; (2) the horizontal fraction is the quotient of the time required to complete one active sweep of the scanning electron beam divided by the time required for both the active and the retrace sweeps; (3) the number of frames per second is usually 30 but higher frame rate systems are sometimes used; (4) the lines per frame is the number of horizontal scanning lines per frame; (5) the aspect ratio is the ratio of the width of a television frame to its height; and (6) the expression is multiplied by 2 because resolution is described in terms of the total number of resolution elements, both light and dark, that are visible in the television image. If the number of horizontal scanning lines per frame is increased, as in a high-resolution television system, then the visibility of detail in the image may be increased in the vertical direction. However, an increase in bandwidth is required if the image detail is to be maintained or improved in the horizontal direction. Most television systems provide an image with about equal horizontal and vertical resolution.

Example 14-4

An image of light and dark vertical and horizontal lines is presented on the output screen of an image intensifier. This image is transmitted to a television monitor by a 525-line television camera. The bandwidth is 3.5 MHz for the television chain. The framing time of the camera is 1/30 second. What is the maximum number of light and dark lines on the output screen that are duplicated without distortion on the television monitor?

With an aspect ratio of 1 and a horizontal fraction of 0.83 (since the retrace sweep is five times as rapid as the active sweep), the horizontal resolution may be computed with Eq. (14-8):

$$\begin{aligned} \text{Horizontal resolution} &= \frac{2[\text{bandwidth}][\text{horizontal fraction}][\text{aspect ratio}]}{(\text{frames/sec})(\text{lines/frame})} \\ &= \frac{2(3.5 \times 10^6 \text{ Hz})(0.83)(1.0)}{(30 \text{ frames/sec})(50 \text{ lines/frame})} \\ &= 370 \text{ lines} \end{aligned}$$

The image will be transmitted undistorted if it is composed of no more than 185 dark lines separated by 185 light lines. As described in Example 14-3, the vertical resolution for a 525-line camera is 343 lines. Hence the vertical and horizontal resolutions are about equal.

TABLE 14-1 Bandwidth Required for Various Television Systems to Provide Equal Resolution in Horizontal and Vertical Directions Across the Image

<i>No. of Scan Lines/Frame</i>	<i>Frames/sec</i>	<i>Vertical and Horizontal Resolution (Lines)</i>	<i>Bandwidth (MHz)</i>
525	30	343	3.3
625	25	408	3.8
837	30	547	8.3
875	30	571	9.1
945	30	617	10.5
1024	16.7	670	6.9

Data in Table 14-1 show the bandwidths required for different television systems to furnish equal resolution in the vertical and horizontal directions. These data were computed for an aspect ratio of 1.0 and a horizontal fraction of 0.83.

DIGITAL FLUOROSCOPY

One technique for digital fluoroscopy is digitization of the signal from the video camera. Other techniques, in which the video camera is replaced by a digital detector such as a CCD camera or a flat-panel detector (see Chapter 13), are becoming increasingly popular and are expected ultimately to replace digitization of the video camera signal.

There are a number of advantages to digital fluoroscopic techniques. First, the digital still-frame image (digital photospot) serves as an alternative to the spot film (taken from a cassette on the input side of the image intensifier) or the analog photospot image (taken on 105-mm film on the output side of the image intensifier). Digital photospots have the advantages of lower patient dose and the possibility of image processing to enhance structures of interest. Second, the digital image may be used for various purposes in real-time fluoroscopy. For example, the most efficient use of radiation is to produce a bright image on the output phosphor with a pulse of radiation, and to use the TV camera to scan every line of the image to produce the full frame as quickly as possible. The digital “scan converter” may be used to store the full digitized frame. The full frame data are then read out and transmitted as two interlaced fields to a standard television monitor. This “pulsed progressive” technique reduces the loss of resolution that occurs in “continuous interlaced” scanning due to motion. Such loss of resolution is particularly troublesome in applications involving rapidly changing structures, such as cardiac catheterization. “Pulsed progressive” scanning results in a substantial (30% to 50%) dose reduction when compared with the standard video method⁹ of scanning every other line to produce a field and repeating the process to produce the interlaced frame, all during continuous x-ray exposure.

Another possible advantage of digital techniques in fluoroscopy is the ability to perform “road mapping,” a digital frame store-and-display mode that retains the most recent fluoroscopic image on the screen as an aid to the placement of catheters and guidewires for interventional procedures. Road mapping employing frame-hold procedures reduces the need for continuous x-ray exposure of the patient. Once multiple digital images are available, image processing techniques such as frame averaging and edge enhancement may be used to improve the data presentation.

Various image processing techniques are used to enhance image quality in digital fluoroscopy and to acquire information that would be difficult to obtain if the images were analog (on cut film or cine film). One example of digital image processing in fluoroscopy is frame averaging. In digital fluoroscopy, images are being acquired rapidly, and the level of quantum noise may be quite high in any one frame. When viewed as multiple frames per second, the eye and brain of the observer do their own “averaging” of quantum noise over several frames so that the noise is suppressed. When viewed as individual frames, as in road mapping, a reduction in quantum noise is desirable. This may be accomplished by averaging corresponding pixel values over

several frames. Over a period of less than a second, the image features may not change significantly, but the quantum fluctuation (quantum noise) does change. By frame averaging, the effects of noise in each pixel may be reduced without significantly altering the pixel value (signal) that represents the attenuation of x rays in the patient.¹⁰

Example 14-5

Find the theoretical improvement in image quality (reduction of noise) if 3, 5, and 7 digital images are averaged.

The digital signal is directly proportional to the number of photons striking the image receptor. Digital noise in a quantum limited system varies as the square root of the number of photons. Therefore, the signal-to-noise ratio (SNR) is

$$\text{SNR} = \frac{\text{Signal}}{\text{Noise}} \propto \frac{N}{\sqrt{N}} = \sqrt{N}$$

and the following table may be constructed:

Images Averaged	SNR
3	1.73
5	2.23
7	2.65

The table shows that averaging 3 frames per second yields an increase in SNR by a factor of 1.73.

Note that the relative improvement diminishes as more frames are averaged. Seven frames is more than twice as many as three frames, but the SNR is not twice its relative value at three frames. This observation, coupled with the fact that averaging more frames is likely to produce more motion blur places a limit, fewer than 10, upon the maximum number of frames that are usually averaged.

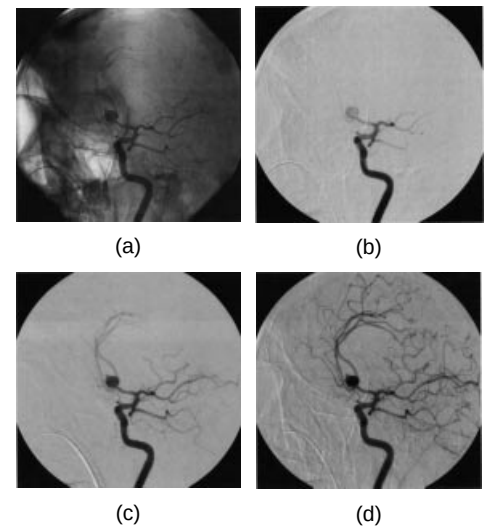
Digital Subtraction Angiography

Images (acquired pre- and postinjection of contrast material) may be subtracted digitally to yield images of the contrast-laden vessels without the distraction of other structures in the patient (Margin Figure 14-7). Typically, a “mask” image is selected before contrast injection. The mask is subtracted from succeeding images after contrast injection (Margin Figure 14-8). If patient motion occurs after the mask has been selected, then anatomical structures in succeeding images will not coincide and artifacts will occur. If this occurs, possible solutions include:

- **Remasking:** If other precontrast images occur after motion has ceased but before contrast injection, a new mask may be selected.
- **Pixel shifting:** If motion is not too complex, the mask digital image may be shifted one or more pixels in all directions until the smoothest image is obtained. An algorithm to accomplish this task without human intervention is available in most digital subtraction angiography systems.

AUTOMATIC BRIGHTNESS CONTROL

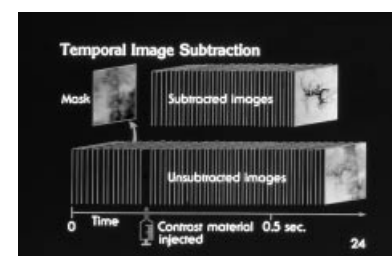
In fluoroscopy, an automatic brightness control (ABC) unit is used to maintain an image of constant brightness on the output screen of the intensifier, irrespective of changes in the attenuation of the x-ray beam in the patient. An image of constant brightness is achieved by automatic adjustment of exposure technique variables (e.g., peak kilovoltage, milliamperes, and x-ray pulse width) in response to fluctuations



MARGIN FIGURE 14-7

Digital subtraction angiography. Cerebral arteriogram (oblique transorbital view) shows an aneurysm of the anterior communicating artery (junction of the A₁ and A₂ segments of the anterior cerebral artery). **A:** Unsubtracted original digital fluoroscopic image obtained midway through the contrast material injection.

B–D: Subtracted DSA images obtained at three progressive time points during contrast material injection. (from ref 10 with permission)



MARGIN FIGURE 14-8

The timing of events in digital subtraction angiography. During serial acquisition, a mask image is selected from images taken prior to injection of contrast material. The mask image is subtracted from images taken after injection of contrast material to obtain the “subtracted” images. From Ritenour ER Computer Applications in Diagnostic Radiology, Mosby, Chicago, 1983.

in brightness of the output screen. For purposes of discussion, an ABC unit may be separated into two components: (1) the brightness-sensing element, with an output used to control (2) the machine's technique variables, which are adjusted automatically to maintain constant image brightness.^{11,12}

Brightness-Sensing Elements

The earliest technique to monitor image brightness was continuous measurement of the image intensifier photocathode–anode current. With this approach there is no simple way to compensate for bright-edge effects at the periphery of the image that are caused by x rays unattenuated by the patient. Compensation for variable collimation usually is provided by a shutter compensation control, which reduces the ABC's response as field size is decreased.

In systems that offer television viewing of the fluoroscopic image, a second approach is available to monitor the brightness of the entire image on the output screen. This approach uses the intensity of the television signal as an indication of output screen brightness. The advantage of this technique is that the portion of the image actually monitored can be selected electronically. If the center of the image is monitored, bright-edge and collimation effects are eliminated.

The most common method of brightness sensing uses a photomultiplier tube to monitor the brightness of the output screen of the intensifier. The area monitored can be defined optically with lenses and diaphragms, thereby eliminating edge and collimation effects. This approach has the advantage of less sensitivity to noise and image intensifier aging effects and, along with the television method, is not influenced greatly by x-ray energy.

Technique Variables Controlled by Automatic Brightness Control

Three variables that can be controlled by an ABC unit are the peak kilovoltage, the tube current, and the width of the x-ray pulses in x-ray units with variable pulse width.

Peak Kilovoltage Variability

With systems of this type, the current (in milliamperes) is selected by a continuous or fixed-step current control, and the peak kilovoltage is selected automatically to provide an image of desired brightness. In the automatic mode, the selected current exerts some influence on the peak kilovoltage required to furnish a satisfactory image, that is, high milliamperage lowers the peak kilovoltage and low milliamperage raises the peak kilovoltage. On some machines, the peak kilovoltage control establishes the maximum peak kilovoltage that the ABC can select. On units with an automatic/manual control, the manual setting disables the ABC unit and permits the operator to select both the milliamperage and the peak kilovoltage. X-ray systems with peak-kilovoltage-variable ABC units usually provide a wider dynamic range of brightness control.

Current Variability

A current-variable ABC control offers the advantage of operator control of peak kilovoltage during ABC operation. At the peak kilovoltage selected, however, the tube current must be sufficiently variable to furnish an image of adequate brightness over a wide range of x-ray beam attenuation. Usually, the dynamic range of a current-variable unit is less than that of a peak-kilovoltage-variable unit. Units with variable current have a relatively long response time because they depend on the thermal characteristics of the x-ray tube filament.

Peak-Kilovoltage-Current Variability

Some ABC units vary both the peak kilovoltage and the current, with both technique variables changed in the same direction (i.e., either increased or decreased). This

approach offers a wide dynamic range but provides little control over the peak kilovoltage selected for the examination. For this reason, some peak kilovoltage/current-variable systems offer the option of manual selection of either peak kilovoltage or current. With these options, operation is identical to that described above for current-variable or peak-kilovoltage-variable units.

Pulse Width Variability

Machines equipped with digital or cine cameras sometimes use grid-controlled x-ray tubes to vary the width (time) of the voltage pulse across the x-ray tube. In this manner, the brightness of the image can be kept constant during fluoroscopy. In most units of this type, the milliamperage is set at a fixed value, and the pulse width of the x-ray beam is varied up to some maximum limit. A peak kilovoltage override circuit increases the kilovoltage above that selected if an image of adequate brightness is not achieved at maximum pulse width. The pulse-width-variable type of control has the distinct advantages of instantaneous response and, with the peak kilovoltage override circuit, a very wide dynamic range.

Combination Circuits

Any combination of the above approaches to ABC operation is possible, but the one most commonly used, other than peak voltage–current variability, is the peak kilovoltage override capability. This method may be applied to either milliamperage-variable or pulse-width-variable machines. The peak kilovoltage override approach provides automatic compensation if the selected peak kilovoltage is not adequate to furnish an image of desired brightness.

CINEFLUOROGRAPHY

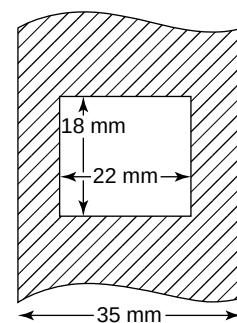
The image on the output screen of an image intensifier may be photographed with a 35-mm movie camera. Although this technique has largely been replaced by digital techniques, it is still performed in a few centers.

Grid-controlled x-ray tubes are used to synchronize the x-ray exposure with the cinefluorographic film. With these x-ray tubes, the bias voltage on the focusing cup surrounding the filament is regulated by a timing circuit that is synchronized with the film-supply mechanism of the cinefluorographic camera. Electrons flow across the x-ray tube only when the cinefluorographic film is in position for exposure.

“Flickering” of the image in the viewing mirror is noticeable during cinefluorography with a synchronized cinefluorographic unit operated at frame rates slower than about 30 frames per second. The rate at which the image flickers depends on the number of frames of film exposed per second.

The photographic film used for cinefluorography must be sensitive to the wavelength of light from the output screen of the image intensifier. Orthochromatic emulsions (sensitive to all visible light except red light) and panchromatic emulsions (sensitive to all wavelengths of visible light) are used for cinefluorography. A relatively high-speed film with high inherent contrast should be used for cinefluorography. The film should be able to record fine detail. No single film satisfies all these requirements, and a compromise must be reached when a particular film is chosen for a cinefluorographic examination.

Each frame of 35-mm film is 22 mm wide and 18 mm long and furnishes 16 frames per linear foot. Each frame of 35-mm film is rectangular rather than square (Margin Figure 14-9). If the size of the recorded image is restricted to the smaller dimension of a frame, then the frame is not exposed at the borders along the larger dimension. Expanding the image to fill the larger dimension of the film is referred to as *overframing*. Overframing causes the loss of part of the image at the borders along the shorter dimension of the frame. If the lost portion of the image is not important clinically, then overframing is useful because it expands the image over a larger area of film and may increase the image detail that is recorded.



MARGIN FIGURE 14-9
Dimensions of single frames of 35-mm cinefluorographic film.

Example 14-6

An image intensifier provides a resolution of 1.5 line pairs per millimeter on an input screen 9 in. in diameter. With overframing, the minification of the image on 35-mm cinefluorographic film is

$$\frac{(9 \text{ in.})(2.54 \text{ cm/in.})}{2.2 \text{ cm}} = 10.4$$

Therefore, the film should provide a resolution of

$$(1.5 \text{ line pairs/mm})(10.4) = 15 \text{ line pairs/mm}$$

Cinefluorographic film is capable of this level of resolution.

During cinefluorography, the current through the x-ray tube is increased many times over that required for routine fluoroscopy with an image intensifier. The increased current is necessary to provide an image on the output screen that is bright enough to expose each frame of cinefluorographic film. This exposure requires an exposure rate of approximately 10 μR per frame at the input screen of the image intensifier.¹³ The exposure rate at the entrance surface of the patient will be 100 to 1000 times this value. Because exposure rate increases linearly with current through the x-ray tube, exposure of the patient to radiation is also increased many times during a cinefluorographic examination.^{14–16} A similar increase in patient dose occurs in digital fluoroscopy, depending upon the number of frames per second and the tube current used.

Photospot Images

Spot photographs of the image on the output screen of an image intensifier may be acquired with 100-mm C-4 film or 105-mm roll film. Photospot images require less exposure of the patient to radiation when compared with spot films.¹⁷ Although the image quality of a photospot image may be slightly less than that furnished by a conventional spot film, photospot images are satisfactory for most examinations.¹⁸

PROBLEMS

- *14-1. An image intensifier has an input screen 9 in. in diameter and an output screen 1 in. in diameter. The flux gain of the intensifier is 40. What is the total gain in luminance?
- 14-2. In an image intensification system, trace the information and its carrier from the x-ray beam emerging from the patient to the visual image on the observer's retina. Describe the factors that contribute to resolution loss at each stage of this transformation.
- *14-3. A patient 12 in. thick is positioned on a table 15 in. from the target of an x-ray tube. The region of interest in the patient is 6 in. in diameter and 3 in. above the table. Can the entire region be displayed at once with an image intensifier with an input screen 9 in. in diameter?
- *14-4. A television system with 875 scan lines per frame provides 815 active sweeps per frame. If the Kell factor is 0.7, the horizontal fraction is 0.83, the aspect ratio is 1.0, and the frame rate is 30 frames per second, what bandwidth is required to provide equal resolution horizontally and vertically?
- 14-5. What is meant by vignetting in an image furnished by image intensification fluoroscopy? What is meant by "pincushion distortion"?
- *14-6. The exposure rate at tabletop for an abdominal fluoroscopic study of a particular patient is $1 \times 10^{-3} \text{ C/(kg-min)}$ in the 16-in. mode. What exposure rate would be expected on the same equipment if switched to the 12-in. mode? 6-in. mode?

*For those problems marked with an asterisk, answers are provided on p. 492.

SUMMARY

- An image intensifier employs both minification gain and electronic (flux) gain.
- Reducing the field of view of an image intensifier results in a magnified image but increases the dose to the patient over the smaller area.
- Standard broadcast television images consist of two interlaced fields that constitute a single frame. Sixty fields appear per second, and therefore there are 30 frames per second.
- Road mapping and frame hold systems can be used to reduce patient dose in fluoroscopy.
- Signal-to-noise ratio varies as the square root of the number of photons. Therefore it increases as patient dose is increased in a given imaging system.
- In digital systems, frame averaging can be used to reduce noise.
- Remasking and pixel shifting are possible remedies for motion artifacts in digital subtraction angiography.

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OBJECTIVES

After completing this chapter, the reader should be able to:

- Explain the principles of x-ray transmission computed tomography.
- Compare the properties of x-ray projection images with x-ray CT images.
- Provide a brief history of the evolution of x-ray CT imaging.
- Describe different approaches to the reconstruction of CT images from projection measurements.
- Depict the configuration of spiral and “ultrafast” CT scanners.
- Outline the features of x-ray sources, detectors, collimators, and display systems used in x-ray CT.
- Characterize the relationship between CT numbers, linear attenuation coefficients, and physical densities associated with CT scanning.
- Identify various characteristics important in quality control of CT units.

INTRODUCTION

In conventional radiography, subtle differences of less than about 5 percent in subject contrast (i.e., x-ray attenuation in the body) are not visible in the image. This limitation exists for the following reasons:

1. The projection of three-dimensional anatomic information onto a two-dimensional image receptor obscures subtle differences in x-ray transmission through structures aligned parallel to the x-ray beam. Although conventional tomography resolves this problem to some degree, structures above and below the tomographic section may remain visible as “ghosts” in the image if they differ significantly in their x-ray attenuating properties from structures in the section.
2. Conventional image receptors (i.e., film, intensifying and fluoroscopic screens) are not able to resolve small differences (e.g., 2%) in the intensity of incident radiation.
3. Large-area x-ray beams used in conventional radiography produce considerable scattered radiation that interferes with the display of subtle differences in subject contrast.

To a significant degree, each of these difficulties is eliminated in computed tomography (CT). Hence, differences of a few tenths of a percent in subject contrast are revealed in the CT image. Although the spatial resolution of a millimeter or so provided by CT is notably poorer than that provided by conventional radiography, the superior visualization of subject contrast, together with the display of anatomy across planes (e.g., cross-sectional) that are not accessible by conventional imaging techniques, make CT exceptionally useful for visualizing anatomy in many regions of the body.

HISTORY

The image reconstruction techniques used in CT were developed for use in radio astronomy,¹ electron microscopy,² and optics including holographic interferometry.^{3–5} In 1961, Oldendorf explored the principle of CT with an apparatus using an ¹³¹I source.⁶ Shortly thereafter, Kuhl and colleagues developed emission and ²⁴¹Am transmission CT imaging systems and described the application of these systems to brain imaging.^{7,8} In spite of these early efforts, CT remained unexploited for clinical imaging until the announcement by EMI Ltd. in 1972 of the first commercially available x-ray transmission CT unit designed exclusively for studies of the head.⁹ The prototype for this unit had been studied since 1970 at Atkinson–Morley

The term “tomography” is derived from the Greek word *tomos*, meaning “section.”

Computed tomography was not the first x-ray method to produce cross-sectional images. In the late 1940s and 1950s, Takahashi in Japan published several papers describing the analog technique of transverse axial tomography. Takahashi's method was commercialized by Toshiba Inc. in the 1960s. The Toshiba product was usurped by computed tomography in the early 1970s.

William Oldendorf was a neuroscientist interested in improving the differentiation of brain tissue. In particular, he was searching for a better imaging method than pneumoencephalography for studying the brain. Many scientists believe he should have shared in the 1970 Nobel Prize in Medicine.

The 1970 Nobel Prize in Medicine was shared by two pioneers of computed tomography, Godfrey Hounsfield, an engineer with EMI Ltd., and Allen Cormack, a South African medical physicist.

Hounsfield was a radar instructor during World War II. Before developing the first x-ray CT scanner, he designed the first transistorized digital computer while working as an engineer at EMI Ltd.

Hospital in England, and the first commercial unit was installed in the United States in 1973. The same year, Ledley and colleagues announced the development of a whole-body CT scanner.¹⁰ In 1974, Ohio Nuclear Inc. also developed a whole-body CT scanner, and clinical models of both units were installed in 1975. By 1977, 16 or so commercial companies were marketing more than 30 models of transmission CT scanners. Today, approximately 5000 CT units are installed in U.S. hospitals, at a cost of up to a million dollars or slightly more per unit.

■ PRINCIPLE OF COMPUTED TOMOGRAPHIC IMAGING

In early CT imaging devices (“scanners”) a narrow x-ray beam is scanned across a patient in synchrony with a radiation detector on the opposite side of the patient. If the beam is monoenergetic or nearly so, the transmission of x rays through the patient is

$$I = I_0 e^{-\mu x}$$

In this equation the patient is assumed to be a homogeneous medium. If the x-ray beam is intercepted by two regions with attenuation coefficients μ_1 and μ_2 and thicknesses x_1 and x_2 , the x-ray transmission is

$$I = I_0 e^{-(\mu_1 x_1 + \mu_2 x_2)}$$

If many (n) regions with different linear attenuation coefficients occur along the path of x-rays, the transmission is

$$I = I_0 e^{-\sum_{i=1}^n \mu_i x_i}$$

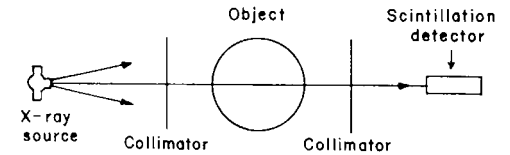
where $-\sum_{i=1}^n \mu_i x_i = (\mu_1 x_1 + \mu_2 x_2 + \cdots + \mu_n x_n)$, and the fractional transmission I/I_0 is

$$e^{-\sum_{i=1}^n \mu_i x_i}$$

With a single transmission measurement, the separate attenuation coefficients cannot be determined because there are too many unknown values of μ_i in the equation. However, with multiple transmission measurements in the same plane but at different orientations of the x-ray source and detector, the coefficients can be separated so that a cross-sectional display of attenuation coefficients is obtained across the plane of transmission measurements. By assigning gray levels to different ranges of attenuation coefficients, a gray-scale image can be produced that represents various structures in the patient with different x-ray attenuation characteristics. This gray-scale display of attenuation coefficients constitutes a CT image.

In early (first-generation) CT scanners, multiple x-ray transmission measurements are obtained by scanning a pencil-like beam of x rays and an NaI detector in a straight line on opposite sides of the patient (Figure 15-1A). During this translational scan of perhaps 40 cm in length, multiple (e.g., 160) measurements of x-ray transmission are obtained. Next, the angular orientation of the scanning device is incremented 1 degree, and a second translational scan of 160 transmission measurements is performed. This process of translational scanning separated by 1-degree increments is repeated through an arc of 180 degrees. In this manner, $160 \times 180 = 28,800$ x-ray transmission measurements are accumulated. These measurements are transmitted to a computer equipped with a mathematical package for reconstructing an image of attenuation coefficients across the anatomic plane defined by the scanning x-ray beam.

Cormack was interested in the generation of cross-sectional displays of attenuation coefficients for use in treatment planning for radiation therapy. He published his results in the *Journal of Applied Physics*. He received only one reprint request. It was from members of the Swiss Avalanche Research Center, who were interested in possible use of his method to predict snow depths.



MARGIN FIGURE 15-1

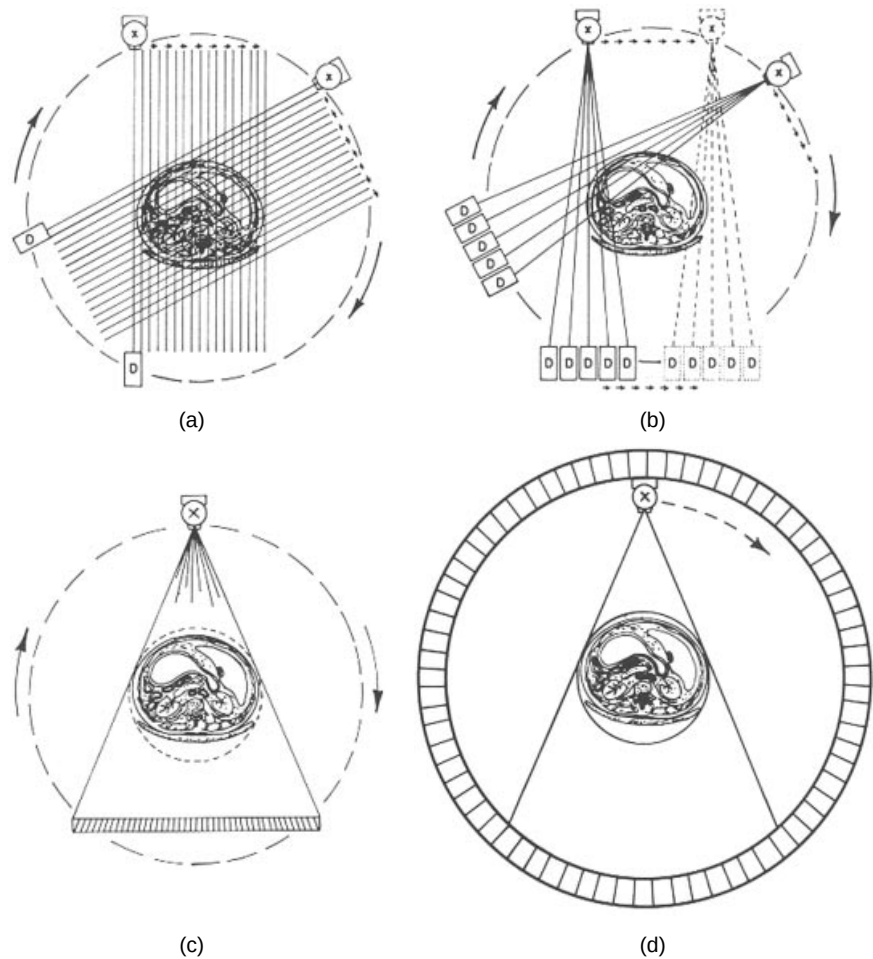
In a first-generation computed tomographic scanner, x-ray transmission measurements are accumulated while an x-ray source and detector are translated and rotated in synchrony on opposite sides of the patient.

Hounsfield termed his technique “computerized transverse axial tomography.” This expression was later abbreviated to “computerized axial tomography,” and was referred to as “CAT scanning.” After sufficient ridicule had been directed toward this acronym, the expression *computed tomography* (CT scanning) was adopted by major journals in medical imaging.

X-ray CT images are often described as “density distributions” because they provide a gray-scale display of linear attenuation coefficients that are closely related to the physical density of tissues.

The scanning technique employed by first generation CT scanners is referred to as “rectilinear pencil-beam scanning.”

The first five clinical CT units marketed by EMI Ltd. were installed by early 1973 in London, Manchester, Glasgow, Rochester (MN), and Boston.

**FIGURE 15-1**

Scan motions in computed tomography. **A:** first-generation scanner using a pencil x-ray beam and a combination of translational and rotational motion. **B:** Second-generation scanner with a fan x-ray beam, multiple detectors, and a combination of translational and rotational motion. **C:** Third-generation scanner using a fan x-ray beam and smooth rotational motion of x-ray tube and detector array. **D:** Fourth-generation scanner with rotational motion of the x-ray tube within a stationary circular array of 600 or more detectors.

Backprojection is also known as the *summation method* or the *linear supposition method*.

Simple backprojection does not produce sharp and clear images, and it is not used in commercial CT scanners.

The integral equations approach to image reconstruction was first described in 1917 by the Austrian mathematician Radon. He developed his equations for studies of gravitational fields.¹²

As used in a mathematical sense, the word *filter* connotes a mathematical operation on a set of data. The convolution filter (also called a *kernel*) discussed here is applied to the data before they are backprojected to form an image.

RECONSTRUCTION ALGORITHMS

The foundation of the mathematical package for image reconstruction is the reconstruction algorithm, which may be one of four types.¹¹

1. *Simple backprojection*. In this method, each x-ray transmission path through the body is divided into equally spaced elements, and each element is assumed to contribute equally to the total attenuation along the x-ray path. By summing the attenuation for each element over all x-ray paths that intersect the element at different angular orientations, a final summed attenuation coefficient is determined for each element. When this coefficient is combined with the summed coefficients for all other elements in the anatomic section scanned by the x-ray beam, a composite image of attenuation coefficients is obtained. Although the simple backprojection approach to reconstruction algorithms is straightforward, it produces blurred images of sharp features in the object.
2. *Filtered backprojection*. This reconstruction algorithm, often referred to as the *convolution method*, uses a one-dimensional integral equation for the

reconstruction of a two-dimensional image. In the convolution method of using integral equations, a deblurring function is combined (convolved) with the x-ray transmission data to remove most of the blurring before the data are backprojected. The most common deblurring function is a filter that removes the frequency components of the x-ray transmission data that are responsible for most of the blurring in the composite image. One of the advantages of the convolution method of image reconstruction is that the image can be constructed while x-ray transmission data are being collected. The convolution method is the most popular reconstruction algorithm used today in CT.

3. *Fourier transform*. In this approach, the x-ray attenuation pattern at each angular orientation is separated into frequency components of various amplitudes, similar to the way a musical note can be divided into relative contributions of different frequencies. From these frequency components, the entire image is assembled in “frequency space” into a spatially correct image and then reconstructed by an inverse Fourier transform process.
4. *Series expansion*. In this technique, variations of which are known as ART (algebraic reconstruction technique), ILST (iterative least-squares technique), and SIRT (simultaneous iterative reconstruction technique), x-ray attenuation data at one angular orientation are divided into equally spaced elements along each of several rays. These data are compared with similar data at a different angular orientations, and differences in x-ray attenuation at the two orientations are added equally to the appropriate elements. This process is repeated for all angular orientations, with a decreasing fraction of the attenuation differences added each time to ensure convergence of the reconstruction data. In this method, all x-ray attenuation data must be available before reconstruction can begin.

A high-frequency convolution filter reduces noise and makes the image appear “smoother.” A low-frequency filter enhances edges and makes the image appear “sharper.” A low-frequency filter may be referred to as a “high-pass” filter because it suppresses low frequencies and allows high frequencies to pass.

Filtered backprojection removes the star-like blurring seen in simple backprojection. It is the principal reconstruction algorithm used in CT scanners.

The Fourier approach to image reconstruction is used commonly in magnetic resonance imaging, but seldom in CT scanning.

In 1958, the Ukrainian physicist Korenblyum¹³ from the Kiev Polytechnic Institute reworked Radon’s integral equations for application to fan-beam geometry. He verified his approach with images of a rotating body captured on film.

Series expansion (iterative reconstruction) techniques are not used in commercial CT scanners because the iteration cannot be started until all of the projection data have been acquired, causing a delay in reconstruction of the image.

The first CT scan of a patient was acquired at London’s Atkinson-Morley Hospital in 1972.

In the mid-1970s more than 20 companies were developing x-ray CT scanners for the clinical market. Today, CT scanners are manufactured only by large international manufacturers of x-ray imaging equipment.

Because of their extended data acquisition and image reconstruction times, first- and second-generation CT scanners were limited principally to studies of the head and extremities where methods to immobilize anatomy could be employed.

Development of a purely rotational CT scanner required a more complex reconstruction algorithm that accommodated a purely rotational geometry. This algorithm was developed in the mid-1970s by scientists at General Electric Medical Systems.

SCAN MOTIONS

First- to Fourth-Generation CT Scanners

Early (*first-generation*) CT scanners used a pencil-like beam of x-rays and a combination of translational and rotational motion to accumulate the many transmission measurements required for image reconstruction (Figure 15-1A). Although this approach yields satisfactory images of stationary objects, considerable time (4 to 5 minutes) is required for data accumulation, and the images are subject to motion blurring. Soon after the introduction of pencil-like beam scanners, fan-shaped x-ray beams were introduced so that multiple measures of x-ray transmission could be made simultaneously (Figure 15-1B). Fan beam geometries with increments of a few degrees for the different angular orientations (e.g., a 30-degree fan beam and 10-degree angular increments) reduced the scan time to 20 to 60 seconds. Fan beam geometries also improved image quality by reducing the effects of motion. CT scanners with x-ray fan beam geometries and multiple radiation detectors constitute the *second generation* of CT scanners.

The third and fourth generations of CT scanners eliminate the translational motion of previous scanners and rely exclusively upon rotational motion of the x-ray tube and detector array (third generation Figure 15-1, C) or upon rotational motion of the x-ray tube within a stationary circular array of 700 or more detectors (fourth-generation scanner, Figure 15-1D). With these scanners, data accumulation times as short as 1 second are achievable.

Spiral CT Scanning

Several approaches to even faster CT scans have been pursued. Until recently, multiple scan sequences to produce contiguous image “slices” required that the x-ray tube stop its rotation and reverse its direction because the maximum extension of the

Spiral CT scanning is also known as *helical CT scanning*.

Spiral CT scanners today employ multiple detector rings to scan several slices through the body during each gantry rotation. These scanners are referred to as multislice CT scanners.

The principal advantage of spiral CT is its ability to image a larger volume of tissue in a relatively short time. With spiral CT, for example, the entire torso can be imaged in a single breathhold.

A pitch of one yields a contiguous spiral. A pitch of two yields an extended spiral. A pitch of $1/2$ yields an overlapping spiral.

Multislice CT scanners have many advantages, but also some disadvantages. Hundreds of CT images can be accumulated in a single study, resulting in high patient doses and massive amounts of digital imaging data.

TABLE 15-1 Advantages of Spiral Compared with Conventional Computed Tomography

• Faster image acquisition • Quicker response to contrast media • Fewer motion artifacts • Improved two-axis resolution • Physiological imaging • Improved coronal, sagittal, and 3D imaging • Less partial volume artifact • No misregistration
--

Research on the dynamic spatial reconstructor has been discontinued.

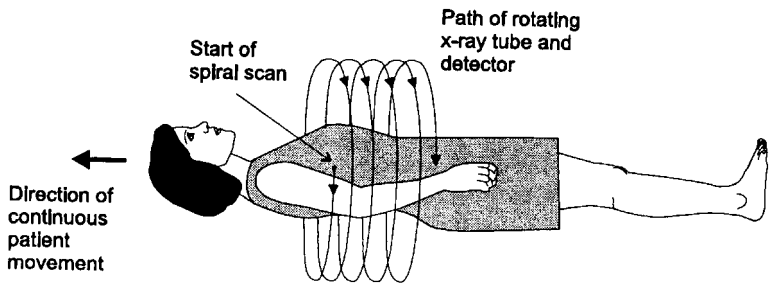


FIGURE 15-2
Spiral CT scanning. (From Rowlands, J.¹⁴ Used with permission.)

high-voltage cables had been reached. Thus, a successive slice-by-slice accumulation technique was used to produced multislice images. In this technique, the total image acquisition time is significantly longer than the beam-on time because the table increments (moves) to the next slice location and the patient breathes between slices.

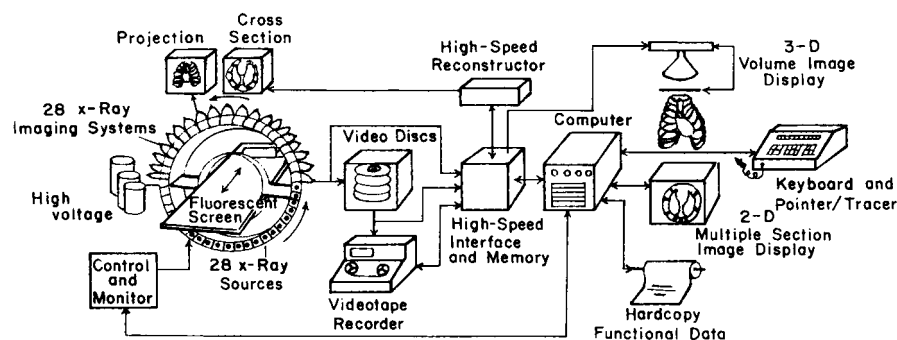
Spiral CT scanning was introduced in 1989 and is used almost universally today for third- and fourth-generation CT scanning. In this approach, image acquisition time is decreased significantly by connecting the tube voltage cables through a “slip ring” or sliding contact mounted on the rotating gantry of the unit. With slip ring technology, the x-ray tube rotates while the patient table moves without stopping. Hence, the patient is moved continuously through the gantry during the study, and the x-ray beam maps out a helical or spiral path in the patient, as depicted in Figure 15-2. Potential advantages of the spiral CT technique include a reduction of patient motion and a general increase in patient throughput.¹⁵ A greater volume of the patient may be scanned during the passage of contrast media, permitting reduction in the volume of contrast needed. Also, the continuity of data along the axis of the patient (i.e., absence of gaps between scans) improves the quality of three-dimensional reconstruction.¹⁶

In single-slice CT scanning, *pitch* is defined as the *patient couch movement per rotation divided by the slice thickness*. In multislice CT, this definition is altered slightly to *patient couch movement per rotation divided by the beam width*. Low pitch (i.e., small increments of couch movement) yields improved spatial resolution along the long axis (Z axis) of the patient, but also results in higher patient doses and longer imaging times. For pitches greater than unity, the dose to the patient is less, but data must be interpolated so that resolution along the Z axis is preserved. Advantages of spiral over sectional computed tomography are listed in Table 15-1.

Ultrafast CT Scanners

Other approaches to fast CT scanning have involved radically different approaches to equipment design. In the late 1970s the first approach to subsecond CT scans was proposed by a group at the Mayo Clinic.¹⁷ This approach, known as the dynamic spatial reconstructor (DSR), incorporated 28 gantry-mounted x-ray tubes over a 180-degree arc and used an equal number of image intensifier assemblies mounted on the opposite semicircle of the gantry. The entire assembly rotated about the patient at a rate of 15 rpm to provide 28 views every 1/60 second. Working models of the system were built for research,¹⁸ but the technical complexity and cost prevented the DSR from being marketed commercially. A diagram of the DSR is shown in Figure 15-3.

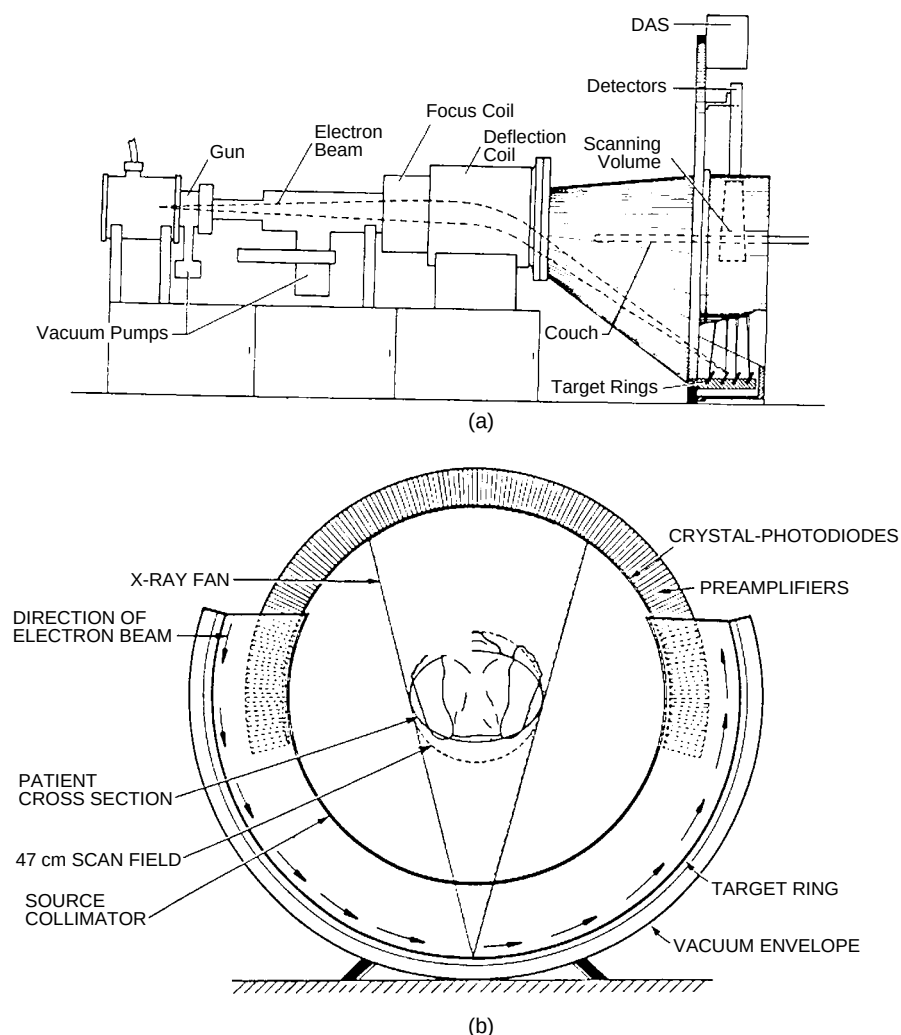
Another approach to fast CT scanning eliminates mechanical motion entirely by converting the gantry of the unit into a giant x-ray tube in which the focal spot moves electronically about the patient.²⁰ This device, known as ultrafast CT (UFCT), cardiovascular CT (CVCT), or “cine CT,” incorporates a semicircular tungsten x-ray target into the gantry.^{21,22} A scanning electron beam with an energy of 130 keV is swept around the semicircular target so that the focal spot moves around the patient. A stationary semicircular bank of detectors records the x-ray transmission in a fashion

**FIGURE 15-3**

The dynamic spatial reconstructor. (From Robb, R., Lent, A., and Chu, A.¹⁹ Used with permission.)

similar to that of a fourth-generation scanner. Because of the speed with which the electron beam may be steered magnetically, a scan may be accomplished in as little as 50 ms and repeated after a delay of 9 ms to yield up to 17 images per second.²³ By using four target rings and two detector banks, eight slices of the patient may be imaged without moving the patient. A diagram of a scanning electron beam CT scanner is shown in Figure 15-4.

The UFCT is often referred to as electron-beam computed tomography [EBCT]. It was initially referred to as a cardiovascular CT scanner.



The UFCT unit was developed in the late 1970s by D. Boyd and colleagues at the University of California—San Francisco.

FIGURE 15-4

Longitudinal (A) and cross-sectional (B) view of the Imatron UFCT scanner.²⁴

In CT units, the heel effect is eliminated by placing the anode–cathode axis of the x-ray tube at right angles to the long axis of the patient.

CT scanners employ compact, high-frequency x-ray generators that are positioned inside the CT gantry. In some units the generator rotates with the x-ray tube, while in others the generator is stationary.

A special x-ray filter is used in CT to make the intensity of the x-ray beam more uniform. This filter is often referred to as “bow-tie filter.”

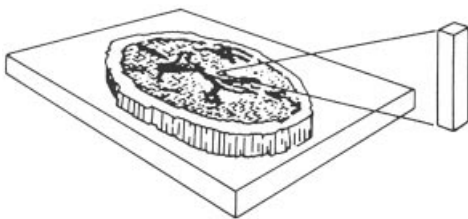
Multislice CT scanning has placed added demands on the capacity of x-ray tubes to sustain high power levels over long periods of time.

The voxel size is the major influence on image resolution in most CT units.

CT units employ two types of collimators: source (prepatient) collimators to shape the x-ray beam and limit patient dose, and detector (postpatient) collimators to control slice thickness.

Reducing CT slice thickness yields the following:

- Decreased partial volume artifact
- Fewer x rays incident upon the detector
- Noisier images



MARGIN FIGURE 15-2

A three-dimensional volume of tissue (voxel) displayed as a two-dimensional element in the CT image (pixel).

High-pressure xenon detectors provide detection efficiencies of about 50%. The detection efficiency of solid-state detectors used in CT is about 80%.

X-RAY SOURCES

Both stationary- and rotating-anode x-ray tubes have been used in CT scanners. Many of the translate–rotate CT scanners have an oil-cooled, stationary-anode x-ray tube with a focal spot on the order of 2×16 mm. The limited output of these x-ray tubes necessitates a sampling time of about 5 ms for each measurement of x-ray transmission. This sampling time, together with the time required to move and rotate the source and detector, limits the speed with which data can be accumulated with CT units using translational and rotational motion.

To reduce the sampling time to 2 to 3 ms, newer CT units use 10,000 rpm rotating-anode x-ray tubes, often with a pulsed x-ray beam, to achieve higher x-ray outputs. To meet the demands of high-speed CT scanning, x-ray tubes with ratings in excess of 6 million heat units are becoming standard.

COLLIMATION

After transmission through the patient, the x-ray beam is collimated to confine the transmission measurement to a slice with a thickness of a few millimeters. Collimation also serves to reduce scattered radiation to less than 1% of the primary beam intensity. The height of the collimator defines the thickness of the CT slice. This height, when combined with the area of a single picture element (pixel) in the display, defines the three-dimensional volume element (voxel) in the patient corresponding to the two-dimensional pixel of the display.

A voxel encompassing a boundary between two tissue structures (e.g., muscle and bone) yields an attenuation coefficient for the pixel that is intermediate between the values for the two structures. This “partial-volume artifact” may be reduced by narrowing the collimator to yield thinner slices. However, this approach reduces the number of x rays incident upon the detector. With fewer x rays interacting in the detector, the resulting signals are subject to greater statistical fluctuation and yield a noisier image in the final display.

X-RAY DETECTORS

To reduce the detector response time, all detectors used in CT are operated in current rather than pulse mode. Also, rejection of scattered radiation is accomplished with detector collimators rather than pulse height analyzers. Detectors for CT scanning, either gas-filled ionization chambers or solid-state detectors, are chosen for their detection efficiency, short response time, and stability of operation. Solid-state detectors include NaI (Tl), CaF₂, and CsI scintillation crystals; ceramic materials containing rare-earth oxides; and bismuth germanate (BGO) and cadmium tungstate [CdWO₄] detectors chosen for their high detection efficiency and low fluorescence decay time. Gas-filled ionization chambers contain xenon pressurized up to 25 atm to improve their x-ray detection efficiency. With any detector, the stability of response from one transmission measurement to the next is essential for production of artifact-free images. With a rotational source and detector geometry, for example, detector instability gives rise to ring-shaped artifacts in the image. Minimum energy dependence of the detectors over the energy range of the x-ray beam is also important if corrections for beam hardening are to be applicable to all patient sizes and configurations.

VIEWING SYSTEMS

The numbers computed by the reconstruction algorithm are not exact values of attenuation coefficients. Instead, they are integers termed CT numbers that are related to attenuation coefficients. On most CT units, the CT numbers range from –1000

TABLE 15-2 Electron Densities of Various Body Tissues^a

Tissue	Electron Density (Electrons/cm ³)	Physical Density (g/cm ³)
Water	3.35×10^{-23}	1.00
Bone	3.72–5.59	1.2–1.8
Spleen	3.52	1.06
Liver	3.51	1.05
Heart	3.46	1.04
Muscle	3.44	1.06
Brain		
White matter	3.42	1.03
Gray matter	3.43	1.04
Kidney	3.42	1.05
Pancreas	3.40	1.02
Fat	3.07	0.92
Lung	0.83	0.25

^aData were calculated from the atomic composition and physical density.

Source: Geise and McCullough.²⁵ Used with permission.

for air to +1000 for bone, with the CT number for water set at 0. The relationship between CT number and linear attenuation coefficient μ of a material is

$$\text{CT number} = 1000 \frac{(\mu - \mu_w)}{\mu_w}$$

where μ_w is the linear attenuation coefficient of water.

CT numbers normalized in this manner provide a range of several CT numbers for a 1% change in attenuation coefficient.

A television monitor is used to portray CT numbers as a gray-scale visual display. This viewing device contains a contrast enhancement feature that superimposes the shades of gray available in the display device (i.e., the dynamic range of the display) over the range of CT numbers of diagnostic interest. Control of image contrast with the contrast enhancement feature is essential in x-ray CT because the electron density, and therefore the x-ray attenuation, are remarkably similar for most tissues of diagnostic interest. This similarity is apparent from the data in Table 15-2. The same cross-sectional CT data displayed at different settings of the “window” of the contrast enhancement control are illustrated in Figure 15-5. The viewing console of the CT scanner may contain auxiliary features such as image magnification, quantitative and statistical data display, and patient identification data. Also, many scanners permit the display of coronal and sagittal images by combining reconstruction data for successive slices through the body.

PATIENT DOSE

The radiation dose delivered during a CT scan is somewhat greater than that administered for an equivalent radiographic image. A CT image of the head requires a dose of about 1 to 2 rad, for example, whereas an abdominal CT image usually requires a dose of 3 to 5 rad. These doses would have to be increased significantly to improve the contrast and spatial resolution of CT images. The relationship between resolution and dose can be approximated as

$$D = a \left(\frac{s^2}{e^3 b} \right) \quad (15-1)$$

where D is the patient dose, s is the signal/noise ratio, e is the spatial resolution, b is the slice thickness, and a is a constant. From Eq. (15-1), the following are apparent:

In a CT image, higher CT numbers are brighter and lower CT numbers are darker.

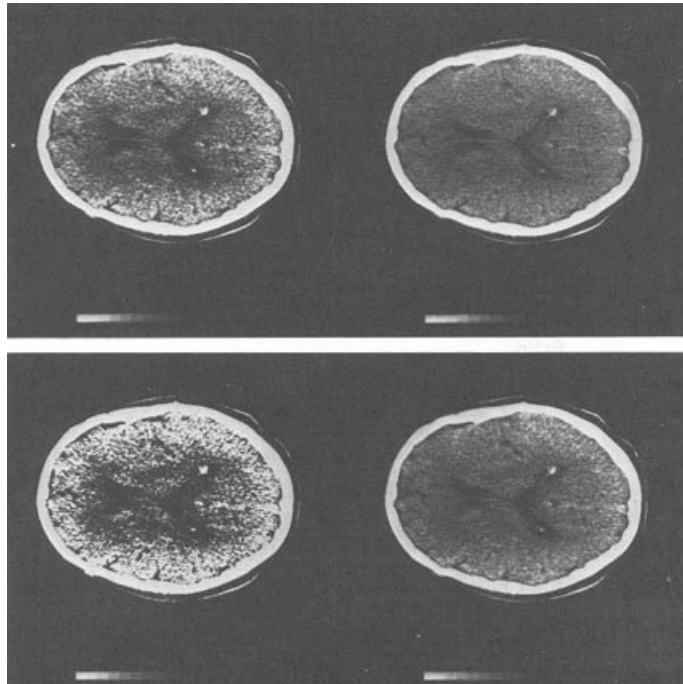
CT numbers are occasionally, but unofficially, referred to as *Hounsfield units*.

Linear attenuation coefficients of various body tissues for 60 keV x rays.²³

Tissue	M (cm ⁻¹)
Bone	0.528
Blood	0.208
Gray matter	0.212
White matter	0.213
CSF	0.207
Water	0.206
Fat	0.185
Air	0.0004

Filtering of lower-energy x rays from the x-ray beam as it penetrates the patient yields a beam of slightly higher energy in the center of the patient. This effect results in reduced attenuation coefficients in the center compared with the periphery. Hence, the center of the image contains pixels of reduced optical density. This effect is known as the “beam-hardening” artifact.

Image storage devices for CT include magnetic tape and disks, digital videotape, and optical disks and tape.

**FIGURE 15-5**

X-ray attenuation data at four positions of the window level of the contrast enhancement control.

1. A twofold improvement in the signal-to-noise ratio (contrast resolution) requires a fourfold increase in patient dose.
2. A twofold improvement in spatial resolution requires an eightfold increase in patient dose.
3. A twofold reduction in slice thickness requires a twofold increase in patient dose.

In multislice computed tomography, patient dose is described as the CT dose index (CTDI). When the distance that the patient moves between slices (the couch increment CI) equals the slice thickness ST, the CTDI equals the dose averaged over all slices (multislice average dose MSAD). When the couch increment is less than the slice thickness, the MSAD is the CTDI multiplied by the ratio of the slice thickness to the couch increment; that is,

$$\text{MSAD} = \text{CTDI} \left[\frac{\text{ST}}{\text{CI}} \right]$$

Patient dose decreases significantly outside of the slice. A conservative rule of thumb (i.e., an overestimate) is that the dose is 1% of the in-slice dose at an axial distance of 10 cm from the slice.

Techniques for measurement of dose parameters in CT have been described in detail by Cacak.²⁶

In projection radiography, the dose is greatest where the x-ray beam enters the patient. In computed tomography the dose is relatively uniform across the section of tissue exposed to the x-ray beam, because the x-ray beam rotates around the patient during exposure.

■ QUALITY CONTROL

Many electronic components and massive amounts of data processing are involved in producing a CT image. A consequence of the separation between data acquisition and image display is the difficulty of observing and investigating imaging system problems through observation of the image alone. In such a complex system, image quality can be ensured only through prospective monitoring of system components and tests of

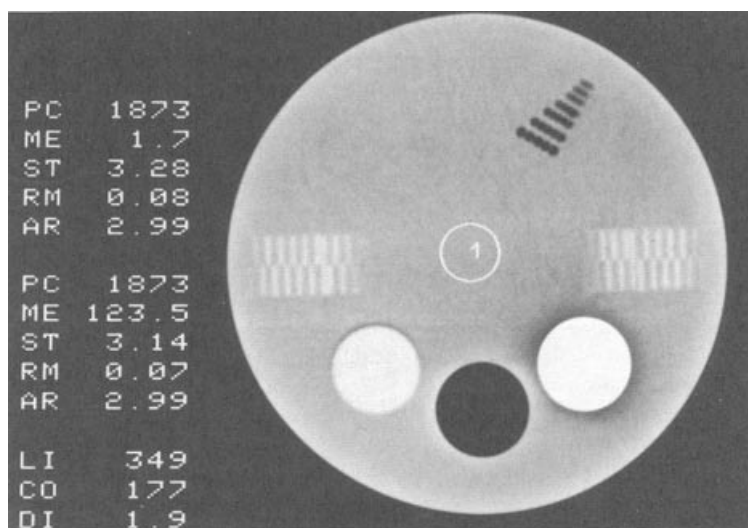


FIGURE 15-6
CT image of a quality control phantom. Image quality is evaluated by analysis of regions of interest and by visual inspection. The mean and standard deviation of pixel values in region 1 indicate CT number calibration, while comparison of region 2 with region 1 yields contrast information. The serrated patterns at 3 and 9 o'clock on the image indicate slice thickness and alignment. The rows of small dark circles (low CT number) at 1 o'clock is an indication of high contrast resolution. (Courtesy of Richard Geise, University of Minnesota Hospital; and Radiation Measurements Incorporated, Middleton, Wisconsin.)

overall system performance with standard phantoms. These measurements should be correlated with patient dose to ensure that the proper balance is maintained among variables that affect contrast, spatial resolution, image noise, and patient radiation dose.

Typical measurements of CT performance are given in Table 15-3, and examples are shown in Figure 15-6. The fundamental system performance indicators are CT number, resolution, noise, and patient dose.²⁶ The accuracy of CT numbers is measured by scanning a water-filled phantom at least monthly. The CT number for water should be zero over a 20-cm-diameter phantom, with a variation of less than 1 CT number. Deviation from the expected CT number of 0 for water at any energy is adjusted by applying a correction factor for the pixel value. Constancy of the value should be monitored with a daily scan.

An overall check of system performance is obtained from semiannual measurements of CT image noise, defined as the standard deviation of CT numbers in a region of interest. Constancy of performance is checked by evaluation of the standard deviation in the daily water scan mentioned previously. Resolution is measured by scanning phantoms on a monthly basis. Of particular importance is low contrast resolution, which is a sensitive indicator of changes in component performance as they affect noise. Patient dose is evaluated semiannually. Specially designed ionization chambers²⁷ provide measurements from which the dose may be calculated for the exposure conditions (narrow beam, variable slice thickness) used in CT.²⁸ The values should agree with manufacturer's specifications to within 20%.

A variety of physical and mechanical factors such as patient couch positioning and indexing should be measured as part of a comprehensive quality control program. The performance of the hard-copy device and system monitors should be checked for distortion, brightness, contrast adjustment, and so on. The accuracy of image analysis features such as distance measurements and measurements of bone density should also be independently evaluated. Additional information on quality control in CT is available in the publications of a number of advisory groups and individuals.²⁹⁻³²

TABLE 15-3 Common Quality Control Measurements for Computed Tomography

Measurement	Frequency
CT number	
Accuracy	Monthly
Constancy check	Daily
Noise	
Evaluation	Semiannually
Constancy check	Daily
Resolution	Monthly
Patient dose	Semiannually

A graph of CT number versus μ should yield a straight line passing through zero for water. This measure, known as the CT number linearity, is essential for quantitative computed tomography.

Reduction in CT number at the center of a water scan is termed "cupping." A heightened CT number at the scan center is known as "peaking." Either result compromises image quality.

SUMMARY

- X-ray transmission CT yields cross-sectional, sagittal, and coronal images with exquisite contrast resolution.
- CT imaging employs the principle of reconstructing images from measurements of x-ray transmission through the body.
- A variety of mathematical models are available for reconstructing x-ray images.
- Special demands are imposed on the x-ray sources and detectors used in CT imaging.
- Quality control and dose limitations are essential features of x-ray CT imaging.

PROBLEMS

- 15-1. Compare the spatial and contrast resolution of conventional radiographic and CT images.
- 15-2. List three reasons why contrast resolution is improved in CT imaging compared with conventional radiographic imaging.
- 15-3. Describe the mechanical features of the four generations of CT scanners.
- 15-4. Briefly describe the simple backprojection approach to image reconstruction, and explain how the convolution method improves these images.
- 15-5. What are the two major constraints on further reductions in CT scan time?
- 15-6. Explain the relationship between a voxel and a pixel.
- 15-7. What types of detectors are used in CT scanners?
- 15-8. Estimate the fetal dose 10 cm from a chest CT examination.
- 15-9. Define a CT (Hounsfield) unit, and explain the purpose of contrast enhancement in a CT viewing device.
- 15-10. Explain how image noise and patient dose in CT scanning are influenced by the signal-to-noise ratio, the size of each resolution element, and the slice thickness.

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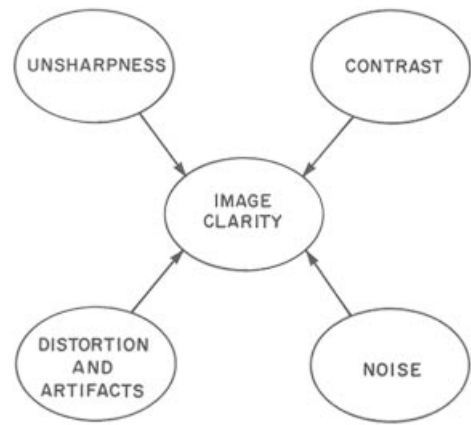
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“A picture may be called a ‘self image’ when it is taken as a visual expression of its own properties, and a ‘likeness’ when it is taken as a statement about other objects, kinds of objects, or properties. The first conception, more elementary, can exist without the second; the second, more sophisticated, combines with the first.”¹ Medical images are used as expressions of the second type of picture, and they combine features of the object with those of the recording process to create (more or less) a ‘likeness’ of the object.

In medical imaging, the word “image” usually connotes a pattern of visible light photons transmitted through a partially transparent film or emitted from a fluorescent screen.²



MARGIN FIGURE 16-1
Contributions to image clarity.

A physician viewing an image with substantial unsharpness might comment that the image lacks detail. The term “detail” is a subjective reaction of the viewer to unsharpness present in an image.

■ OBJECTIVES

After completing this chapter, the reader should be able to:

- Define the meaning of image unsharpness, and identify and explain the variables that contribute to it.
- Define the meaning of image contrast, and identify and explain the variables that contribute to it.
- Define the meaning of image noise, and identify and explain the variables that contribute to it.
- Provide examples of image distortion and artifacts.

■ INTRODUCTION

The purpose of the medical image may be to convey information that contributes to (1) detection of a disease or injury; (2) delineation of the nature and extent of disease or injury; (3) diagnosis of the underlying cause of the disease or injury; (4) guidance of treatment of the disease or injury; and (5) monitoring the treatment and its consequences to determine its effectiveness. The degree to which the image achieves its purpose, or demonstrates that no disease or injury is present, is described by the vague term *image quality*. In part, image quality connotes how clearly the image displays information about the anatomy, physiology, and functional capacity of the patient, including alterations in these characteristics caused by disease or injury. This component of image quality is referred to as the *clarity* of information in the image. Image quality depends on other features also, including whether the proper areas of the patient are examined, whether the correct images are obtained, and even whether a disease or injury is detectable by imaging.

Image clarity is a measure of how well information of interest is displayed in an image. Clarity is influenced by four fundamental characteristics of the image: unsharpness, contrast, noise, and distortion and artifacts. In any image, the clarity of information is affected by these image properties and how they interact with each other.^{3,4}

■ UNSHARPNESS

Every image introduces an element of “fuzziness” or blurring to well-defined (sharp) boundaries in the object (patient). This blurring is described as *unsharpness* and is depicted in Figure 16-1. Unsharpness (termed “blur” in some texts) is a measurable characteristic of images.

Unsharpness in an image is a consequence of four factors that contribute to image formation. These factors, considered collectively as the components of unsharpness, are geometric unsharpness U_g , subject unsharpness U_s , motion unsharpness U_m , and receptor unsharpness U_r . Under conditions where each of these factors contributes independently to the overall unsharpness of the image, the total unsharpness is

$$U_t = [(U_g)^2 + (U_s)^2 + (U_m)^2 + (U_r)^2]^{1/2}$$

Geometric Unsharpness

Geometric unsharpness is a direct consequence of the geometry of the image-forming process. In general, geometric unsharpness is influenced by the size of the radiation source and the distances between source and object (patient) and between object